

Example 9.2.15

A combination lock requires 3 selections of numbers 1 - 30.

- How many combinations? 30^3
- How many without repeats? $30 * 29 * 28$

Example 9.2.27

```
for i = 5 to 50
  for j = 10 to 20
    myFunction();
```

How many function calls? 506.

Example 9.2.3

- How many ways to rearrange the letters in Algorithm? $\frac{9!}{6!}$

9.3 Counting Elements of Disjoint Sets: The Addition Rule

0.1 The Addition Rule

Let $\{A_1, A_2, \dots, A_k\}$ be a partition of A , then $N(A) = N(A_1) + N(A_2) + \dots, N(A_k)$

0.1.1 Example 9.3.7

In some state all license plates consist of 4-6 symbols chosen from 26 letters and 10 digits.

- How many license plates without repetition? $36^4 + 36^5 + 36^6$
- No repetition? $\frac{36!}{32!} + \frac{36!}{31!} + \frac{36!}{30!}$
- How many license plates have repetition? $36^4 + 36^5 + 36^6 - [\frac{36!}{32!} + \frac{36!}{31!} + \frac{36!}{30!}]$
- What is the probability that a random license plate has a repeat?

$$\frac{36^4 + 36^5 + 36^6 - [\frac{36!}{32!} + \frac{36!}{31!} + \frac{36!}{30!}]}{36^4 + 36^5 + 36^6} \quad (1)$$

0.2 The Difference Rule

If $B \subset A \Rightarrow N(A - B) = N(A) - N(B)$

Example 9.3.9b

```

for i = 0 to 4
    for j = 1 to i
        printf("Hello, world");

```

How many runs? $\frac{n(n+1)}{2}$.

0.2.1 Probability of the Complement of an Event

If S is a finite sample space and A is an event in S , then:

$$P(A^c) = 1 - P(A)$$

Let $A = \text{Everyone Aces Exam III}$,
 $A^c = \text{not everyone aces exam III}$.

$$P(A) = 1\%$$

$$P(A^c) = 1 - 1\% = 99\%$$

Assuming all years have 365 days and all birthdays occur with equal probability, how large must n be so that in any randomly chosen group of n people, the probability that two or more have the same birthday is $\geq \frac{1}{2}$

- Probability that no two people have same birthday = $\frac{365 \times 364 \times \dots \times 365 - n + 1}{365^n}$
- Probability that two or more have same birthday = $1 - \frac{365 \times 364 \times \dots \times 365 - n + 1}{365^n}$